

SHRINIRAJ E.K.R

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CAREER OBJECTIVE

Aspiring Embedded Systems and Model-Based Development engineer with a strong passion for Electric Vehicle sub-systems, Battery Management Systems (BMS), and IoT. Currently working as Production Associate Trainee at Electra EV. Seeking an opportunity to gain hands-on experience in Embedded Systems.

EDUCATION

- **2023 - 2027** — Coimbatore
  Karpagam College Of Engineering — B.E- Electrical & Electronics Engineering — **Pursuing (6.5 CGPA)**
- **2021 - 2023** — Komarapalayem
  Royal International School — Higher Secondary Education — **64.6%**
- **2021 - 2021** — Erode
  Bharathi Vidhya Bhavan Higher Secondary School — Secondary Education — **Pass**

HARD SKILLS

Programming

- Basics of C Programming

Technical

- Embedded Systems
- Communication Protocols (UART, I2C, SPI)
- Microcontrollers

SOFT SKILLS & HOBBIES

SOFT SKILLS

- Problem-Solving & Analytical Thinking Team
- Collaboration & Cross-Functional
- Communication Adaptability & Quick Learning in Industrial Environments Time Management & Goal-Oriented Approach

HOBBIES

- Singing
- Reading Books
- Browsing current affairs of my field

LANGUAGES

- German: A1 Elementary Level (In Progress | 515+ Day Streak on Duolingo)
- English: Professional Proficiency
- Tamil: Native/Bilingual Proficiency
- Telugu: Native Proficiency

Technical Training & Coursework

Advanced Diploma Training in Embedded Systems | Cranes Varsity, Bangalore

- Key Modules: Embedded C, Microcontroller Architectures, and Hardware Protocols (UART, I2C, SPI).
- Hands-on work on LPC1768 microcontroller with other components and done some mini projects.

Summer Internship Training Program | Anna University, Chennai (May 2026 – Present)

- Research and training in advanced structural materials and composites. Worked on project, “Edge AI-Managed Bidirectional EV Charging Hub With Boron Nitride Thermal Packaging and Graphene EMI Shielding”.

WORK EXPERIENCE

Trainee - Production Associate at Electra EV

- Performed mechanical module tightening, cell-in-box positioning, and precision torque assembly for BESS enclosures.
- Integrated low-voltage wire harnesses and verified tap-line connectivity, voltage tapping, and signal continuity using BMS Tap Tools.
- Loaded raw operational test logs into OEM diagnostic software to analyze real-time battery parameters and evaluate pass/fail criteria.
- Assisted in short-cycle test data entry and parameter verification for commercial Battery Energy Storage System (BESS).

CERTIFICATIONS

Hackathons & Challenges

- One of the **Top 10 Finalist of International Youth Entrepreneurship Competition 2026 at Ming Chi University of Technology, Taiwan** for the project **Battery Passport System for Used EVs**
- Participated in **"Theeruvathon 2024" conducted by PSG College of Arts and Science, Coimbatore** on project, **Smart Bin Monitoring System.**
- Participated in "**Automation and Robotics Challenge 2025**" by **SRM Institute of Science and Technology, Chennai**
- Participated in "**Embedathon 2025**" conducted by **NIT- K Suratkal**.
- Participated in "**Embedded Programming Challenge 2026**" by **Shaastra 2026 - IIT Madras**

Workshop & Courses

- Completed **NPTEL** course on **Introduction to Industry 4.0 and Industrial Internet of Things** by **IIT Kharagpur (Elite + Gold medal Rank with Top 1%)**
- Completed **MATLAB Onramp** courses on **MATLAB, Simulink, Stateflow, Simscape and Simscape Battery**.
- **Completed** course on **"Battery & Battery Management System (BMS)" by DIYguru E-Mobility Academy.**
- Attended Workshop on **"Electric Vehicle Essentials: Technology, Trends and Future Directions"** conducted by **Sri Ramakrishna Engineering College, Coimbatore**
- Attended Workshop on "**Model-Based Development: From EV powertrains to ADAS**" by **Cloudkampus**.
- Attended Workshop on "**Introduction to ECU and AUTOSAR Architecture for Automotive Systems"** by Skillync

PROJECTS

Smart Bin Monitoring System

- Built an IoT-enabled smart bin that uses sensors to monitor waste levels and automatically alerts collectors when thresholds are reached using ESP 32.
- Implemented real-time notifications and location-based guidance, directing users to nearby bins when the closest bin is full.

Battery Passport System for Used EVs (Participated on IYEC 2026 & Innovent by TATA Technologies)

- Provides a secure digital identity for electric vehicle batteries.
- Records lifecycle data such as charging patterns, health status, usage history, and ownership.
- Integrates with the vehicle's battery system and cloud platforms.
- Creates a tamper-proof record that remains with the battery even if the vehicle is sold or repurposed.

Digital Bank Vault System (Training Project)

- Developed a secure access system using the LPC1768 (ARM Cortex-M3), implementing real-time monitoring of peripheral sensors via GPIO and ADC.
- Engineered a safety logic to trigger alarms and lock mechanisms upon detecting abnormal temperature fluctuations, ensuring physical hardware security.
- Integrated and optimized drivers for I2C, SPI-based sensors and status indicators to provide a responsive user interface and reliable system feedback.

Smart Sensor Fusion based Navigation System (Participated in TATA ELXSI Teleport)

- Designed a navigation logic that fuses data from IMU (Inertial Measurement Unit), odometry, and GNSS to maintain accurate vehicle tracking in signal-challenged environments Raspberry PI 4 Model B.
- Implemented basic filtering algorithms (Complementary filters) to minimize cumulative errors in dead reckoning, improving position estimation for off-highway vehicles.

DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief.